

The relation set of Simplified-V1.Syntactics

Clifford-Relations.Base._SRel, _==_ together with the structural rules of Circuit.Base.Lift-Relation.

Diagrams read left to right in matrix-multiplication order (the *rightmost* gate is applied first), so each transcribes its Agda word letter for letter. Wires are numbered from the bottom: w sits on wire 0, $w \uparrow$ on wire 1, and $w \downarrow$ is w itself.

Derived words (definitions)

$$Z \quad \text{---} \boxed{Z} \text{---} \equiv \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S^{p-1}} \text{---}$$

$$X \quad \text{---} \boxed{X} \text{---} \equiv \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S^{p-1}} \text{---} \boxed{H} \text{---}$$

$$R \quad \text{---} \boxed{R} \text{---} \equiv \text{---} \boxed{S} \text{---} \boxed{Z^{1/2}} \text{---}$$

$$M_x \quad \text{---} \boxed{M_x} \text{---} \equiv \text{---} \boxed{R^x} \text{---} \boxed{H} \text{---} \boxed{R^{x^{-1}}} \text{---} \boxed{H} \text{---} \boxed{R^x} \text{---} \boxed{H} \text{---} \quad x \neq 0$$

$$\begin{array}{c} \text{---} \boxed{\top} \text{---} \\ \perp \top \\ \text{---} \boxed{\perp} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \bullet \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{H} \text{---} \\ | \\ \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{H} \text{---} \bullet \text{---} \end{array}$$

$$\begin{array}{c} \text{---} \boxed{\top} \text{---} \\ \top \perp \\ \text{---} \boxed{\perp} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{H} \text{---} \bullet \text{---} \text{---} \\ | \\ \text{---} \bullet \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{H} \text{---} \end{array}$$

One-wire axioms

$$\text{order-S} \quad \text{---} \boxed{S^p} \text{---} \equiv \text{---}$$

$$\text{order-H} \quad \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \equiv \text{---} \boxed{M_{-1}} \text{---}$$

$$\text{M-power} \quad \text{---} \boxed{(M_g)^k} \text{---} \equiv \text{---} \boxed{M_{g^k}} \text{---}$$

$$\text{semi-MR} \quad \text{---} \boxed{M_g} \text{---} \boxed{R} \text{---} \equiv \text{---} \boxed{R^{g^2}} \text{---} \boxed{M_g} \text{---}$$

$$\text{order-SH} \quad \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \equiv \text{---}$$

$$\text{comm-HHSHHS} \quad \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \equiv \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{H} \text{---}$$

Two-wire axioms

semi-MCZ

semi-MCZ

M_g

$\equiv$

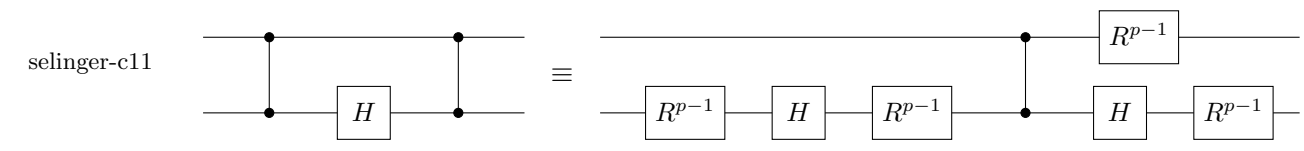
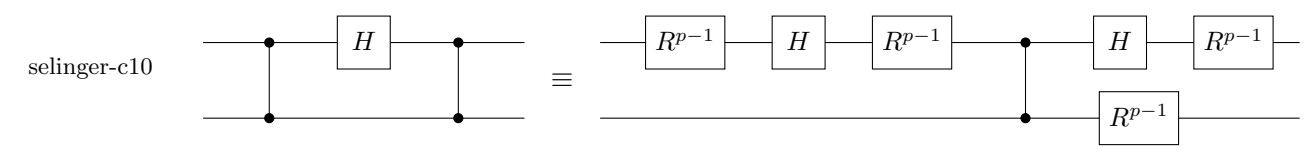
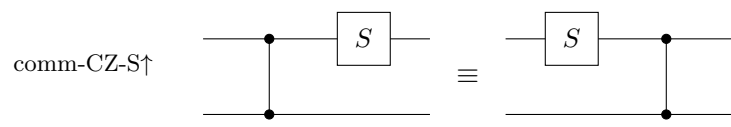
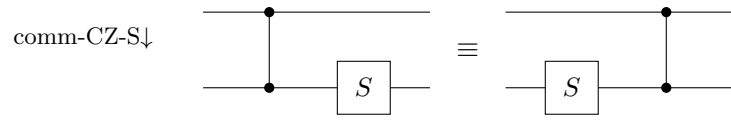
g

M_g

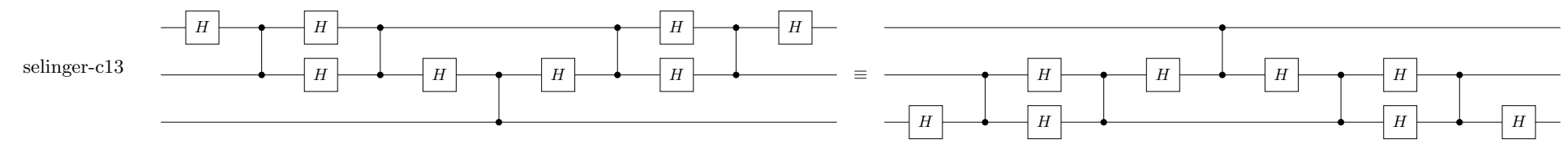
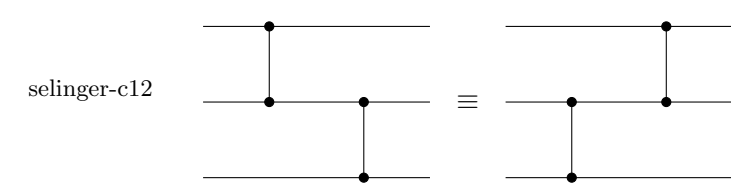
rel-X↓-CZ

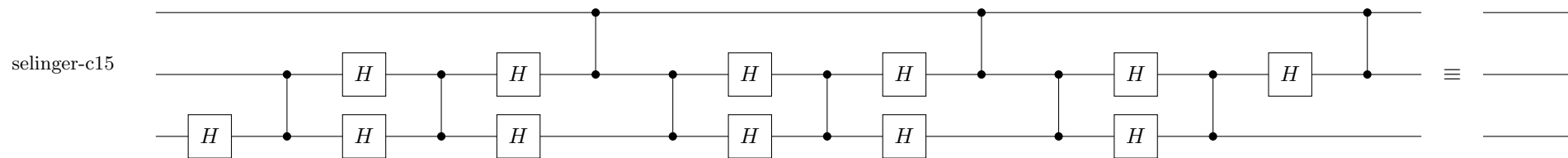
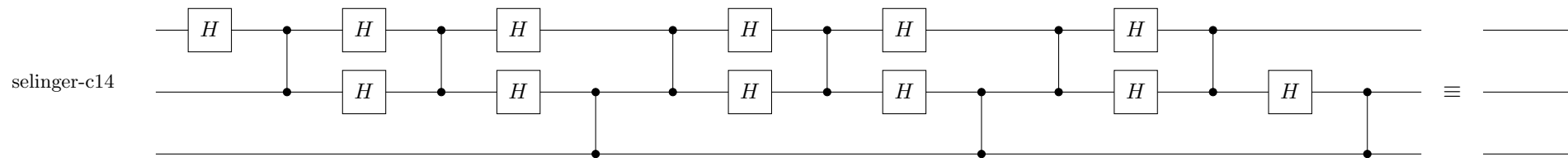
Diagram illustrating the equivalence between two quantum circuit representations for the rel-X↓-CZ gate. The left circuit shows a control line (top) and a target line (bottom). The target line contains a sequence of gates: H , S , H , H , S^{p-1} , and H . The right circuit shows the same gate sequence on the target line, but with the control line (top) and target line (bottom) swapped, and the gates applied to the control line instead.

order-CZ  $\equiv$ 



Three-wire axioms





Structural rules (Circuit.Base.Lift-Relation)

